

EMPLOYMENT

Graduate Research Assistant	NC State University	April 2025 – Present
- Designed a cloud-based genomic data platform with JBrowse 2, enabling interactive exploration of strawberry and grape datasets of 500GB for 100+ researchers worldwide, deployed on Google Cloud Run. - Collaborated on ML architecture design, applying hyperparameter tuning and time-series analysis, improving model accuracy by 12%.		
Product Engineer Intern	Isomer AI	May 2025 – Present
- Owned full backend of a production AI platform — Temporal.io durable workflow orchestration, real-time event pipelines, and complete data layer (TypeScript, Node.js, Express, Prisma ORM, PostgreSQL, Redis, Auth0, Docker, Terraform, Sentry); shipped 10+ features from DB schema to async worker to API across 7 deployments. - Engineered multi-model LLM pipelines across Claude, GPT-4o, and Gemini via MCP with provider-agnostic interfaces — any model swappable without downstream changes; built document processing (PDF gen, email parsing, S3), Playwright test coverage, and reviewed PRs across the engineering team.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML; improved claim accuracy by 25%. - Automated deployments on AWS & GCP, integrated monitoring dashboards with Grafana, and supported model validation and red-teaming to enhance reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led end-to-end SDLC for a data-driven system automating support ticket analysis, reducing processing time by 15%. - Implemented distributed data pipelines using Hadoop + Spark for batch processing, and integrated Kafka streaming for real-time data ingestion, reducing ticket-processing latency by 20%.		

EDUCATION

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Data Science, Object Oriented Development, Software Engineering, Algorithms.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
- Bachelor of Technology in Computer Science; CGPA: 9.32/10 - Courses: Data Warehousing, Big Data Infra, DS & Algorithms, OS, Machine Learning, Software Engineering, Blockchain.		

PROJECTS

LLM Eval Harness	Independent Project	2025
- Built a vendor-neutral benchmark comparing Claude, GPT-4o, and Gemini across 2,000 examples and 3 task types (grounded QA, multi-hop reasoning, claim verification) with 3 repeated runs per model. - Tracked hallucination rate, p50/p95 latency, and cost per call; applied bootstrap significance testing and McNemar test for pairwise model comparisons across 6 evaluation metrics.		
PulseRank – Real-Time Leaderboard	Independent Project	2026
- Engineered an event-driven platform with MySQL outbox pattern, MongoDB TTL projections, and Redis sorted sets for sub-millisecond leaderboard reads; validated multi-store consistency with Testcontainers integration tests. - Shipped full observability with OpenTelemetry traces and Prometheus metrics on all hot paths; benchmarked with k6 and deployed via GitHub Actions CI/CD.		
VectorLift – Semantic Search Engine	Independent Project	2026
- Built a production-grade semantic search system on MS MARCO evaluating 6 pipeline configurations: BM25 baseline (Elasticsearch), dense bi-encoder retrieval fine-tuned with in-batch negatives, and cross-encoder re-ranker. - Evaluated with NDCG@10, MRR@10, and MAP with paired bootstrap significance testing; served via FastAPI with per-stage latency tracking and Kafka-driven corpus indexing into Elasticsearch and Qdrant.		
Demand Paging – Xinu OS	Systems Programming	Oct 2024 – Dec 2024
- Engineered an 8-store virtual memory system in C for Xinu OS with Second-Chance page replacement; reduced page faults by 25%. - Added 7+ kernel system calls, increasing memory performance by 15% using inverted page tables and a custom fault handler.		

LANGUAGES AND TECHNOLOGIES

- Languages:** Python, TypeScript, Java, JavaScript, SQL, C, Bash, R
- AI/ML:** Anthropic Claude API, OpenAI API, Gemini API, LangChain, MCP, Hugging Face, TensorFlow, PyTorch, RAG, Pandas, NumPy
- Cloud & DevOps:** AWS, GCP, Docker, Kubernetes, Terraform, CI/CD pipelines, Grafana, Prometheus, OpenTelemetry, Distributed Systems
- Frameworks/Tools:** Node.js, React, NestJS, Next.js, Temporal.io, Kafka, Spark, Hadoop, Spring Boot, FastAPI, Elasticsearch, Redis, PostgreSQL

ADDITIONAL EXPERIENCE AND AWARDS

- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; developed 3 complete MVPs within 48-hour builds.

- **IEEE Published:** Co-authored paper on Computer Vision-Based Cybersecurity Threat Detection at Springer icSoftComp 2023; presented research reaching 500+ attendees.